Assignment: Image Enhancement + Restoration

Tasks

1. Build, train, and evaluate three CNN-based image classifiers on the CIFAR-10 dataset using the following convolutional modules:
    - Standard (regular) convolution
    - Atrous (dilated) convolution
    - Selective Kernel (SK) convolution
2. Compare the classification performance of the three architectures in terms of:
    - Classification accuracy
    - Precision, recall, and F1-score
    - Number of trainable parameters
    - Computational complexity
    - Training and inference time
3. Build, train, and evaluate a MIRNet (Multi-scale Residual Network) for image restoration and enhancement.
4. Evaluate the effectiveness of MIRNet by applying it to low-quality images that have first been enhanced to super-resolution using your previously trained:
    - SRCNN
    - ESPCN
    - SRResNet
    - SRGAN
    - EDSR
5. Compare the image quality before and after MIRNet restoration using both quantitative metrics (e.g., PSNR, SSIM, and LPIPS) and qualitative visual analysis.
6. Analyze and discuss the performance of the different super-resolution pipelines:
    - SRCNN → MIRNet
    - ESPCN → MIRNet
    - SRResNet → MIRNet
    - SRGAN → MIRNet
    - EDSR → MIRNet
7. Determine which combination produces the best overall image quality and justify your conclusion using experimental results and visual comparisons.